

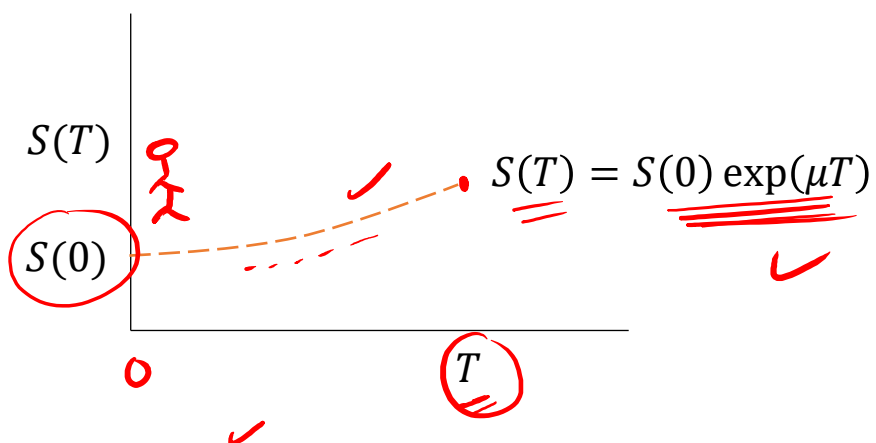
Volatility Models

(1. Introduction to Volatility Models)

What does volatility do ?

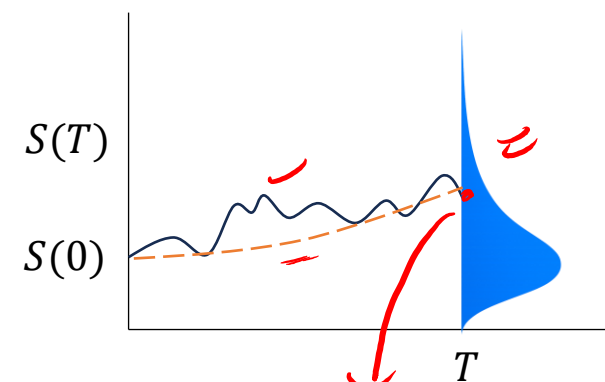
ODE $\frac{dS(t)}{S(t)} = \mu dt$ ✓

$S(T) = S(0) \exp(\mu T)$ ✓



SDE $\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t)$ $dW \sim N(0, dt)$

$S(T) = S(0) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)T + \sigma Z\sqrt{T}\right)$ $Z \sim N(0, 1)$



$\mathbb{E}[S_T]$
 $= S_0 e^{\mu T}$

Motivation

Why volatility model ? If volatility was always constant, we only need to estimate it, not model it. One of the drawbacks of Black-Scholes is that it assumes constant volatility. However, when one backs out volatilities from European Option prices using Black-Scholes, it's different for different strikes and maturities. In other words, the implied volatility (IV) is not a single number.

$$\sigma^{imp} = BSM^{-1}(V, S_0, r, K, T) \quad (\text{Newton Raphson})$$

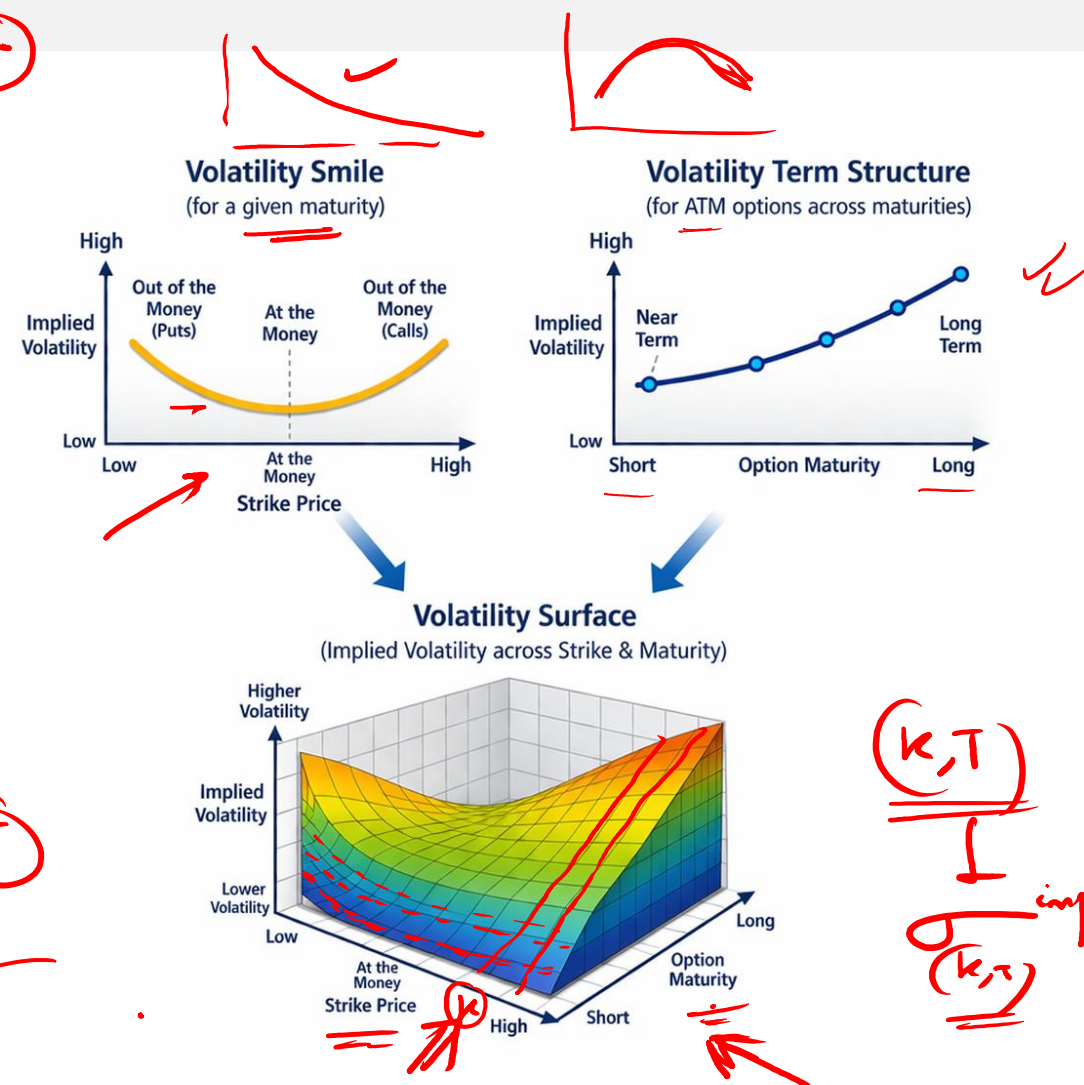
(Handwritten notes: $K \rightarrow \text{fix}$, $T \rightarrow \text{fix}$)

The variation of IV w.r.t different maturities T is called the volatility term structure.

The variation of IV w.r.t different strikes K is called the volatility smile. *(strike dimension)*

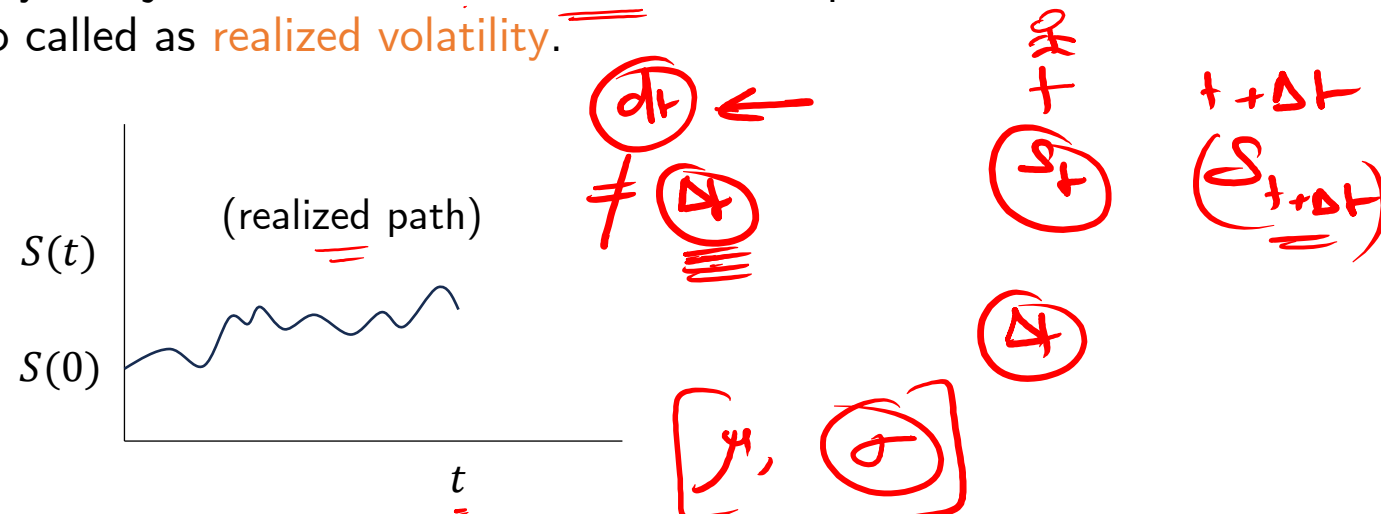
Together it's called as a volatility surface.

σ



Constant vol (estimation)

If the stock price follows constant volatility, we just need to estimate them from past data. The same volatility will prevail in future. The past volatility is also called as realized volatility.



dF →

$$F = (\log S)$$

SDE $\frac{dS(t)}{S(t)} = \mu dt + \sigma dW(t)$ ✓

SDE $d(\log S(t)) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW(t)$ ✓

Discretize

$$S(t + \Delta t) = S(t) \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma Z\sqrt{\Delta t}\right)$$

Discretize

$$\log S(t + \Delta t) = \log S(t) + \left(\mu - \frac{1}{2}\sigma^2\right)\Delta t + \sigma Z\sqrt{\Delta t}$$

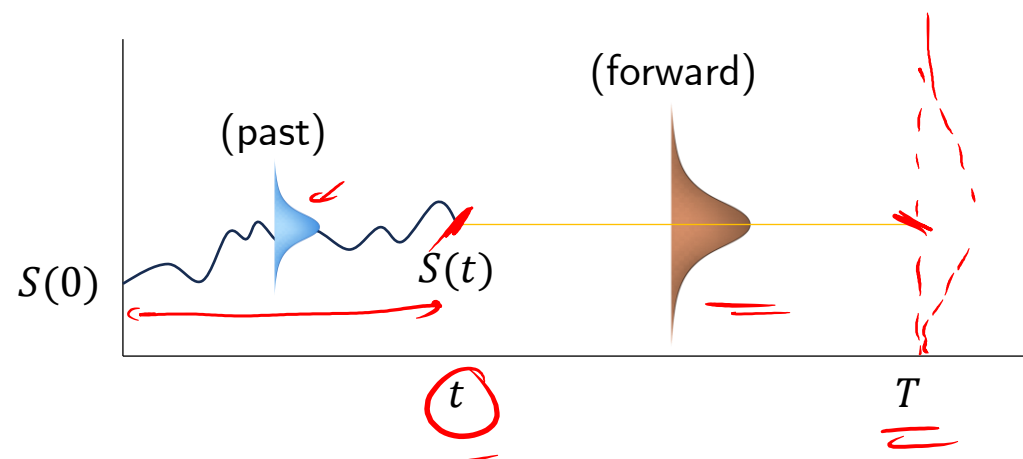
$$\log S(t + \Delta t) - \log S(t) \sim N\left(\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t, \sigma^2\Delta t\right)$$

Option Pricing - Historical vol or Forward vol ?

Which volatility makes more sense from option pricing perspective ? Historical or Forward ? The option price depends on the terminal distribution of the stock price which will depend on the current conditions (S_t) and the future evolution, not the past.

Handwritten diagram illustrating a tree structure:

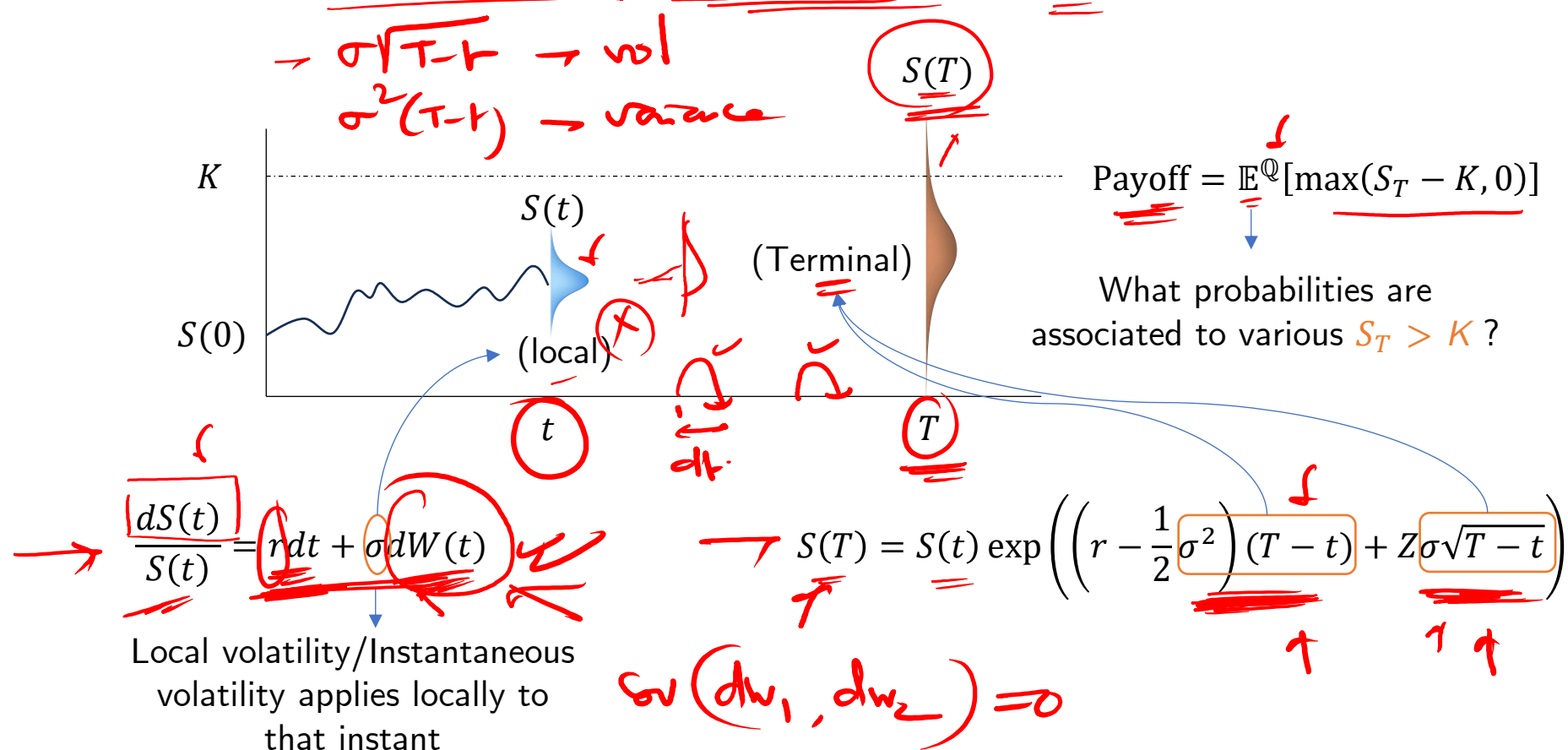
- Root node: T (circled)
- Child node: $GARCH$ (circled)
- Child node of $GARCH$: ω, α, β (circled)



In other words, we can say that the option price packs the information about **forward volatility**

Option Pricing - Local vol or Aggregate vol ?

European option price depends on the terminal density of the stock. The terminal density requires the aggregate volatility between now and maturity. The terminal variance = sum(local variances). Why?



Black-Scholes Price

How is volatility reflected in Black-Scholes price ?

$$\underline{C} = S_t N(d_1) - K e^{-r(T-t)} N(d_2)$$

$$\underline{d_1} = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

$$\underline{d_2} = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$$

σ does not appear in isolation. But it appears only as the aggregated version.

Modeling thoughts ?

What can I do to generate any list of observed option prices/IMPLIED vols ? (**calibration**)

$\sigma \rightarrow$ random / stochastic
 $\rightarrow$ Black, Heston, Hull & White $\oplus$ $\$$

Call Option Prices

	1 Month	3 Months	6 Months
Strike	70	8.81	13.83
70	13.79	13.81	13.83
80	8.82	9.14	9.33
90	4.85	5.74	6.38
100	1.86	3.18	4.11
110	0.47	1.23	1.92
120	0.09	0.35	0.81

$$\sigma^{imp} = \underline{BSM}^{-1}$$

monotonic

Implied Volatilities

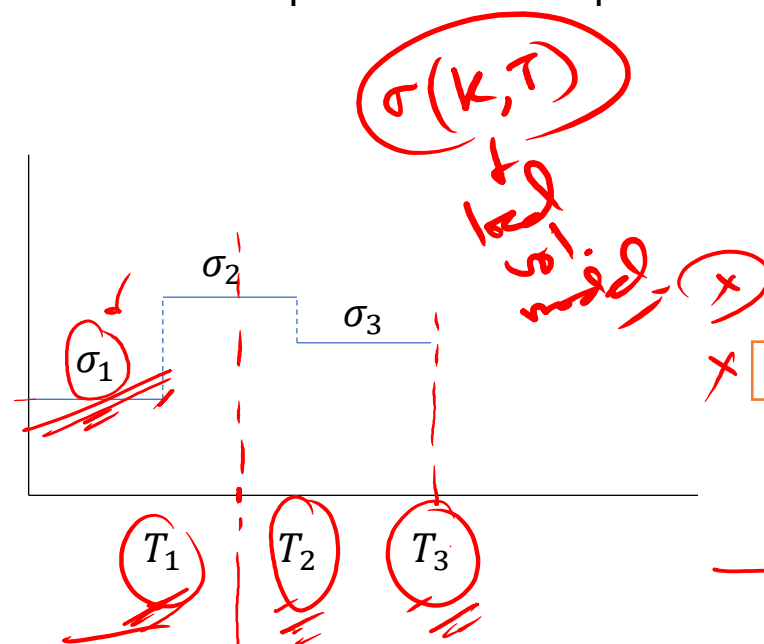
	1 Month	3 Months	6 Months
Strike	30.1%	29.6%	29.2%
70	30.1%	29.6%	29.2%
80	28.7%	28.3%	28.1%
90	26.5%	26.4%	26.5%
100	24.9%	24.8%	25.0%
110	24.4%	24.3%	24.8%
120	24.6%	24.6%	25.1%

$\rightarrow \frac{dS(t)}{S(t)} = rdt + \sigma dW(t)$
 $\sigma(S, t)$
 $\sigma(t), \sigma(S)$

Of course, constant does not work. What else might work ?

Deterministic models - Time dependent vol

If I use a time dependent vol in the GBM, I can fit the term structure side of the surface but not the smile dimension. Since, there are only a discrete set of prices, we will need some interpolation scheme to generate a continuum of prices. The simplest one is piece-wise constant. We could also do piecewise linear, cubic splines etc.



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120	24.6%	24.6%	25.1%

$$\sigma_1^{imp} = \sigma_1$$

$$\sigma_2^{imp} = \frac{1}{\sqrt{T_2}} \sqrt{\sigma_1^2 T_1 + \sigma_2^2 (T_2 - T_1)}$$

$$\sigma_3^{imp} = \frac{1}{\sqrt{T_3}} \sqrt{\sigma_1^2 T_1 + \sigma_2^2 (T_2 - T_1) + \sigma_3^2 (T_3 - T_2)}$$

$$\sigma(t) = a + bt$$

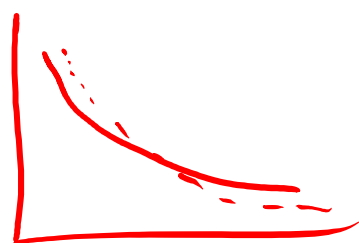
Homework: Try piecewise linear fitting

$$ds/s = rdt + \sigma(t)dw$$

$$\frac{T \times \sigma^{imp}(T)^2}{\int_0^T \sigma(s)^2 ds} = \frac{\int_0^T \sigma(s)^2 ds}{\int_0^T \sigma(s)^2 ds}$$

Deterministic models - State dependent vol

We can use a class of models named as CEV (Constant elasticity of Variance) models where volatility depends on state S .



$$dS(t) = \mu S(t)dt + \sigma S(t)^\beta dW(t)$$

$$ds = \mu s dt + \sigma dw$$

$$\frac{dS(t)}{S(t)} = \mu dt + \underbrace{\sigma S(t)^{\beta-1}}_{\text{(state dependent vol)}} dW(t)$$

$$\mu dt + \left(\sigma \cdot \frac{1}{S^{\beta-1}} \right) dw$$

Note that $\beta = 1$, reduces the models to Black-Scholes

$\beta = 0$ reduces the model to Bachelier model

$\beta < 1$ (used in practice) – if stock price falls, volatility rises (leverage effect => realistic)

$$\Delta S = S (\mu \Delta t + \sigma S^{\beta-1} \cdot z \sqrt{\Delta t})$$

$$S_{t+\Delta t} = S_t (1 + r \Delta t + \sigma S_t^{\beta-1} \cdot z \sqrt{\Delta t})$$

(Euler-Maruyama
discretization)

By calibrating β we will be able to capture some smile dynamics.